

1. Challenge Tackled

Wren addresses the challenge of providing local business owners (such as agricultural hosts and small tour providers) with automated, agentic online presence management. Key aspects of the system include:

- **Agentic Website Creation & Maintenance:** Automatically generating structured, multi-section web pages taking bookings, handling visitor questions, and receiving reviews.
- **Local data collection:** Collecting vendor information in a local language to gather host profile data directly on-device/offline.
- **Automated Q&A Matching & Sentiment Analysis:** Answering visitor inquiries locally in the browser via semantic matching and analyzing visitor review sentiment.

2. Tools / ML Models Used

- **Language Models:** Gemma 4 E2B (general agent runtime and core reasoning), Gemma 4 E4B, Q4 quantized GGUF weights on local server setups.
- **Translation Pipeline:** Dedicated translation model service (NLLB 600M / Kiswahili in-out) running behind explicit translator API endpoints.
- **Speech Recognition (STT):** w2v-BERT 2.0 Kiswahili fine-tune (badrex) for accurate speech-to-text processing during interviews.
- **Text-To-Speech (TTS) & Voice Synthetic:** MMS-TTS and ElevenLabs Eleven v3 (synthetic interviewees) alongside local TTS models.
- **Browser ML & Frameworks:** Transformers.js for client-side vector-based Q&A matching in the user's browser, React Native / Expo app frontend.

3. What Worked Well

- **JSON Schema-Constrained Website Generation:** Switching from freeform markdown generation to explicit English JSON schemas improved page generation reliability from 0/10 to 10/10 valid structured pages.
- **On-Device Browser Q&A:** Client-side Q&A matching via Transformers.js provided instant responses to visitor queries directly in-browser.
- **Decoupled Architecture:** Running the main agent strictly in English while translating user inputs and outputs via dedicated microservices kept logic clean and reliable.

4. What Was Challenging

- **Direct Multilingual Generation Failure:** Directly generating pages with unconstrained LLMs produced gibberish and broken markdown fences, necessitating a structured English-first + translation pipeline.
- **Scope Management & MVP Pivot:** WhatsApp integration posed friction and scope bloat; removing WhatsApp simplified the MVP focus toward automated website creation and sentiment tracking.
- **Offline Audio & STT Benchmarking:** Balancing accuracy against performance across various local speech models required extensive offline testing across Whisper, MMS-1B, and w2v-BERT models.

5. How You Spent Your Time

- **40% — Core Agent & Schema Engineering:** Defining structured JSON output formats, configuring server model endpoints, and resolving test failures.
- **30% — Voice & Translation Stack Setup:** Benchmarking Kiswahili STT/TTS options, setting up NLLB 600M and w2v-BERT models for local execution.
- **20% — Frontend App & Browser Extensions:** Integrating client-side Transformers.js for visitor questions and updating UI components.
- **10% — Architecture Refactoring & Docs Sync:** Pruning cut features (WhatsApp) and maintaining updated project documentation.

6. Next Steps & Future Work

- Finalize the open Farm profile schema (address, per-service availability mapping).
- Implement Google Apps Script / Spreadsheet API backend for storing incoming questions and review logs.
- Implement Whatsapp bot for two-way conversations with customers

Expand local voice interview models for additional regional languages.